

The data from this project comes from a website called Basketball Reference (<https://www.basketball-reference.com/>) which contains a look of free public information about NBA players and teams. The data that was specifically used for the sake of this project can be found under the 'Seasons' tab with the data being the 'Per Game' statistics along with the 'Advanced' statistics for both the 2024-25 and 2025-26 NBA seasons.

I have decided to utilize PostgreSQL for the database to store my project data. I have two reasons as to why I selected PostgreSQL. First is familiarity since I have completed past work with no significant issues. The other reason is that based on the data being used in this project, a relational database makes more sense than using graph databases like Neo4j.

Every year the NBA has a group of players that seem to suddenly improve significantly surprising both teams and fans of the league. While it is easy to predict when some NBA players will breakout into being one of the best players in the NBA, other players can take 8-10 years of development before they achieve their breakout. However, as things currently stand it is difficult for both NBA organizations and fans of the game to confidently predict if a given player will break out that season. This lack of consistency and statistical evidence has led teams to give up on players too early believing that they will never live up to their potential. So when one of these players that told to leave the organization the next season break out into being one of the 50 best players in the NBA this causes executives to lose their jobs, organizations to lose revenue, and can cost a team their chance to win their first ever championship. With this background in mind, I wanted to see if I could utilize NBA statistics from the previous season and machine learning models to accurately classify if a player was going to break out the following year.

My current hypothesis for this project is that players that see an increase by 20% to their points scored per game as well as their estimated win shares will break out in the following NBA season. It is important to mention that I utilized AI to help me refine my hypothesis by asking it to reflect on my hypothesis. ChatGPT highlighted the issue of using a hypothesis that uses only one statistic like points per game can be biased since players that go from averaging 3 to 6 points are not the break out players I was targeting. My final comment regarding my hypothesis is that I am open to refining the details of my hypothesis through feedback from both my instructor and fellow classmates.

I knew that this experiment would be a classification problem which is important to understand before developing a plan for experimentation. While I had some ideas and thoughts for the experiment, I also decided to use AI to help me plan my experiment. The plan below is a combination of my ideas and the plan provided to be my ChatGPT. My plan can be broken down into a total of 3 steps. The first step will be creating the binary label or target variable based on the conditions of my hypothesis. Using my current hypothesis, I would then label players that saw an increase to both points per game and win shares by 20% as class 1 with the rest of the players being labeled under class 0. The next step would be to conduct feature engineering before actually running any machine learning models. This will be done by using insights gained

from data exploration as well as my own knowledge of the NBA and basketball statistics. The final part of my plan is to run all of the machine learning models I decide upon using to compare their results through performance metrics and visualizations. Through the results of my classification models, I should be able to either accept or reject my experiments hypothesis.

Regarding the machine learning models I plan too tentatively use I believe it will be around 3 to 5 classification models depending on the final feature transformations and model performance. These models are logistic regression, decision tree, random forest, gradient boosted trees, and xgboost. I am open to feedback about both the number of models I should run as well as if there are any additional models that I should consider.

The last topic to discuss in this project report is the exploration of data I conducted which can be found under my class GitHub. When it came to handling missing values, I decided to remove the awards column since a majority of the fields were null values. With the remaining missing values I decided to use `fillna(0)` because if the rebounds stat for a player is blank, it does not make sense to just put the median rebound value instead of just setting the value to zero. I used `info()`, `describe()`, and `dtypes()` to look at summary statistics for my dataset. Other than observing that most of the columns in this dataset were numeric in nature, the main discovery I made was that for almost all these columns the range of values fall between 0 to 100 with the outlier being `MPTotal`. I plotted histogram distributions for all the columns along with 3 additional scatterplots. I mainly utilized the code that was provided in the `Session1_In_class_practice` notebook found on canvas. When examining the histogram distributions, it was interesting to observe that all the distributions followed one of two patterns since they either the common centralized bell curve or leaned heavily to the left in terms of value distribution.

I did not use AI to write any of the SQL code, data exploration code, or used to generate the paragraphs of text found here inside of this project report. AI was used to help develop and refine my hypothesis/experiment plan which is mentioned in their corresponding sections.